

Rolling Updates and Rollbacks

These features are provided by a **Deployment** in Kubernetes.

Remember:

Deployment

|



ReplicaSet

|



Pods

|



Containers

A **Deployment** can:

-  Create Pods
-  Scale Pods
-  Update Pods (Rolling Update)
-  Restore an older version (Rollback)

What is a Rolling Update?

Simple Definition

A **Rolling Update** is a method of updating an application without stopping it. Kubernetes replaces old Pods with new Pods one by one.

Why Do We Need Rolling Updates?

Imagine you have a website with **4 Pods**.

Current Version:

Pod 1 → Version 1

Pod 2 → Version 1

Pod 3 → Version 1

Pod 4 → Version 1

Now you want to update your application to **Version 2**.

If you stop all Pods at once:

✗ Users cannot access the website.

This is called **downtime**.

Solution: Rolling Update

Instead of stopping all Pods together,

Kubernetes updates them **one at a time**.

Example

Before Update

Pod 1 → v1

Pod 2 → v1

Pod 3 → v1

Pod 4 → v1

Step 1

Kubernetes creates one new Pod.

Pod 1 → v2 

Pod 2 → v1

Pod 3 → v1

Pod 4 → v1

Step 2

Updates another Pod.

Pod 1 → v2

Pod 2 → v2 

Pod 3 → v1

Pod 4 → v1

Step 3

Pod 1 → v2

Pod 2 → v2

Pod 3 → v2 

Pod 4 → v1

Step 4

Pod 1 → v2

Pod 2 → v2

Pod 3 → v2

Pod 4 → v2 

Now all Pods are updated.

Users never experience downtime.

Real-Life Example

Imagine changing the tires on a bus.

 Wrong way:

Remove all four tires together.

The bus cannot move.

 Correct way:

Change one tire at a time.

The bus keeps moving.

This is exactly how a **Rolling Update** works.

Advantages of Rolling Updates

- No downtime.
 - Users can continue using the application.
 - Safe deployment.
 - Easy monitoring during updates.
 - Easy recovery if something goes wrong.
-

Kubernetes Command for Rolling Update

Suppose you want to update Nginx.

Current version:

nginx:1.24

Update to:

nginx:1.25

Command:

```
kubectl set image deployment/nginx-deployment nginx=nginx:1.25
```

Check progress:

```
kubectl rollout status deployment nginx-deployment
```

What is Rollback?

Simple Definition

Rollback means returning the application to the previous working version if the new update has a problem.

Why Do We Need Rollback?

Suppose you updated:

Version 1

|



Version 2

After the update,

Users report:

- Website is not opening.
- Login is failing.
- Errors are appearing.

The new version has bugs.

Solution

Return to the previous version.

This is called a **Rollback**.

Example

Before Update

Version 1 

↓

Rolling Update

Version 2 

↓

Rollback

Back to Version 1 

Kubernetes Command for Rollback

Undo the last update:

```
kubectl rollout undo deployment nginx-deployment
```

Kubernetes restores the previous version automatically.

View Rollout History

```
kubectl rollout history deployment nginx-deployment
```

Example Output

Revision 1

Revision 2

Check Rollout Status

kubectl rollout status deployment nginx-deployment

Shows whether the update is complete.

Real-Life Example

Think about a mobile app.

You install a new update.

After updating:

- App crashes.
- Buttons don't work.
- Login fails.

You uninstall the update and install the previous version.

That is a **Rollback**.

Rolling Update vs Rollback

Rolling Update

Updates the application

Replaces old Pods with new Pods

Used when releasing a new version

No downtime

Rollback

Restores the previous version

Replaces new Pods with old Pods

Used when the new version fails

Restores a stable version

Complete Flow

Version 1

|



Rolling Update

|



Version 2

|

| If successful



Continue Running

OR

Version 2

|



Problem Found

|



Rollback

|



Version 1

Important Commands

Update Image

kubectl set image deployment/nginx-deployment nginx=nginx:1.25

Updates the application.

Check Update Status

kubectl rollout status deployment nginx-deployment

Shows whether the update is complete.

View Rollout History

kubectl rollout history deployment nginx-deployment

Shows previous deployment revisions.

Rollback

kubectl rollout undo deployment nginx-deployment

Returns to the previous working version.

Quick Revision

Rolling Update

- Updates Pods one by one.
- No downtime.
- Users can continue using the application.
- Performed by a Deployment.

Rollback

- Restores the previous version.
 - Used if the new version has issues.
 - Performed by a Deployment.
-

Easy Memory Trick ★

Version 1

|



Rolling Update

|



Version 2

|

|— Works  → Continue

|

|— Fails  → Rollback

|



Version 1

Interview Definition

Rolling Update:

A deployment strategy where Kubernetes updates Pods gradually, one at a time, ensuring the application remains available with little or no downtime.

Rollback:

The process of reverting a Deployment to a previous stable version when a new deployment causes problems.

One-Line Memory Trick

-  Rolling Update = Gradually update Pods without downtime.
-  Rollback = Go back to the last working version if the update fails.